

Object Oriented Programming (CSC166)

Lab Sheet 5

1. Create a class Employee that stores the employee ID, name, and basic salary. Derive a class Salary that stores House Rent Allowance (HRA), Dearness Allowance (DA), and Bonus of basic salary. Initialize 20 % , 10% and 5% of basic to HRA, DA and bonus. Calculate the gross salary by adding the basic salary, HRA, DA, and bonus, and apply 3% tax deduction on gross salary to compute net salary. Display the employee details along with the salary details.
2. Create a Student with data members such as student ID, name, and semester, initialized through a parameterized constructor. Derive a class Marks that stores marks in five subjects using its own parameterized constructor. Further derive a class Result that calculates the total, percentage, and division (Distinction, First, Second, Pass, or Fail). Assign the division as Distinction ($\geq 80\%$), First (60–79 %), Second (45–59%), Pass (32%), or Fail , if the percentage is below 32% or the student scores less than 32 marks in any subject. Display the complete student details, marks, total, percentage, and division.
3. Create a base class Product with product ID, product name, and unit price initialized using a constructor. Derive a class Order that stores quantity purchased and calculates the total amount. Further derive a class Discount that applies a discount according to the following rules:
 - i. Total $\geq$ Rs. 50,000 $\rightarrow$ 15% discount
 - ii. Total $\geq$ Rs. 20,000 $\rightarrow$ 10% discount
 - iii. Otherwise $\rightarrow$ No discountUse constructors to initialize all required data members and display the final payable amount